

Linear Models, Errors, and One-Sided Tests

AGRON 5130

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Introduction

The goal of this assignment is to help you connect the linear additive model to hypothesis testing, Type I and Type II errors, and one-sided tests.

In this module, we moved beyond formulas and looked at what a statistical test is actually doing underneath the surface. We explored the linear additive model:

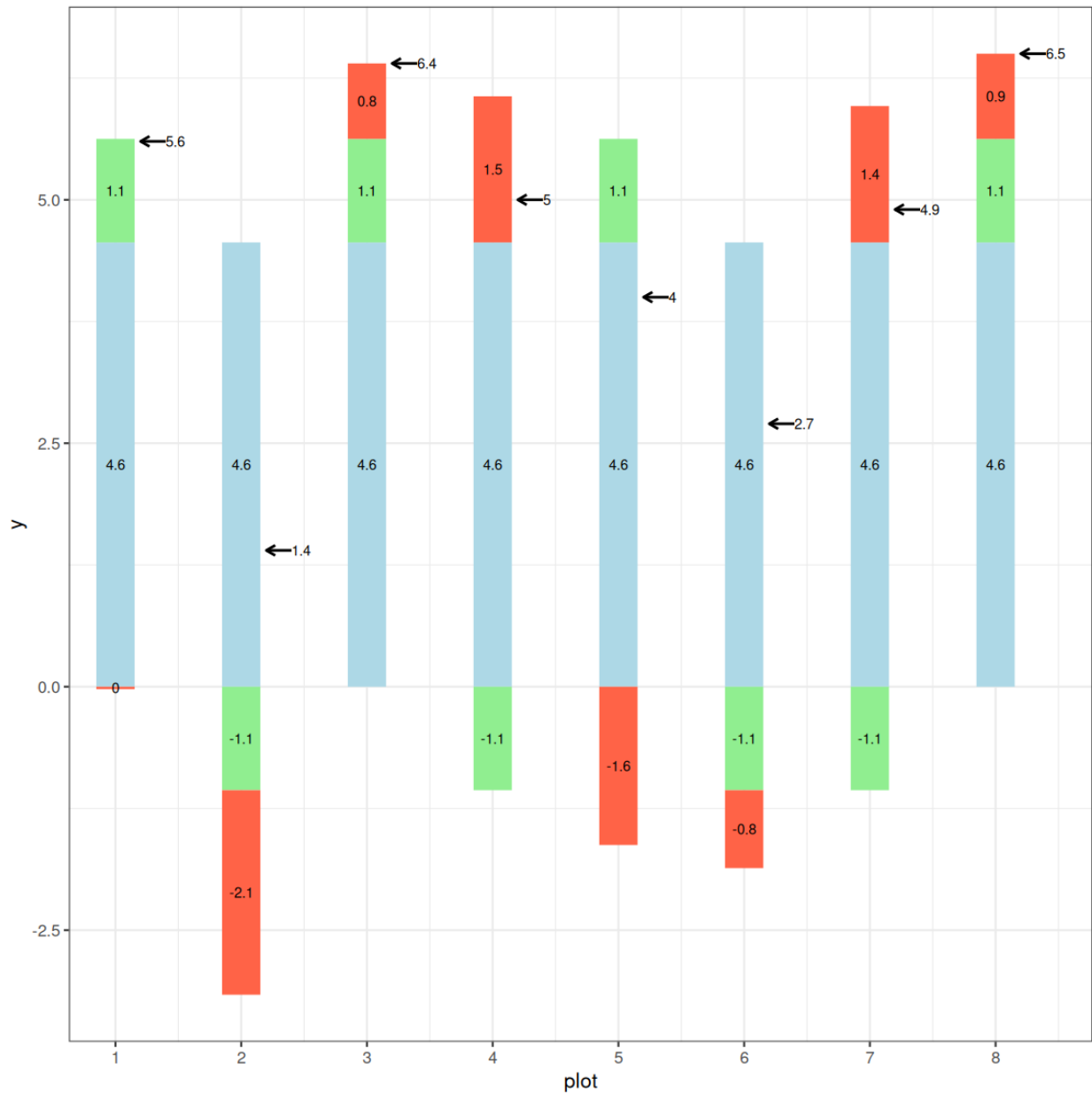
$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

and discussed how treatment effects and random error combine to produce the observed value.

In this assignment, I ask you to interpret model components, think carefully about decision-making under uncertainty, and conduct a one-sided t -test.

Submit a knitted PDF/HTML generated from your R Markdown file to Canvas.

The image below represents the strawberry dataset in a stacked bar plot showing the population mean, treatment effect, and error effect for each plot.



note: When I ask for models and hypotheses, you don't have to spend much time trying to figure out how to write mathematical symbols in R. You can write things in plain text, as long as I can understand what you meant.

You can use the `strawberry` dataset, which already contains the computed effects.

```
strawberry <- read.csv('./data/strawberry_effects_table.csv')
strawberry
```

##	block	trt	y	plot	mu	trt_mean	trt_effect	error_effect	trt_bar_min
## 1	B4	F	5.6	1	4.5625	5.625	1.0625	-0.025	4.5625
## 2	B4	R1	1.4	2	4.5625	3.500	-1.0625	-2.100	-1.0625
## 3	B3	F	6.4	3	4.5625	5.625	1.0625	0.775	4.5625
## 4	B3	R1	5.0	4	4.5625	3.500	-1.0625	1.500	-1.0625

## 5	B2	F	4.0	5	4.5625	5.625	1.0625	-1.625	4.5625
## 6	B2	R1	2.7	6	4.5625	3.500	-1.0625	-0.800	-1.0625
## 7	B1	R1	4.9	7	4.5625	3.500	-1.0625	1.400	-1.0625
## 8	B1	F	6.5	8	4.5625	5.625	1.0625	0.875	4.5625
##	trt_bar_max	err_bar_min	err_bar_max						
## 1	5.625	-0.0250	0.0000						
## 2	0.000	-3.1625	-1.0625						
## 3	5.625	5.6250	6.4000						
## 4	0.000	4.5625	6.0625						
## 5	5.625	-1.6250	0.0000						
## 6	0.000	-1.8625	-1.0625						
## 7	0.000	4.5625	5.9625						
## 8	5.625	5.6250	6.5000						

Question 1

2 points

Given the graph above, write the linear model for the observed value in plot 3, as we did in the exercise.

$$6.4 = 4.5625 + 1.0625 + 0.775$$

Question 2

2 points

Given the plot above, write the linear model for the observed value in plot 5, as we did in the exercise.

$$4.0 = 4.5625 - 1.0625 + 1.5$$

Question 3

2 points

You are testing a new fungicide to control Northern Corn Leaf Blight. Given it also has the potential to improve plant health under heat and drought situations, you plan to charge a price of \$20 per acre. If there is no improvement with the fungicide, the farmer will lose \$20 per acre. Given this high price, would you be more concerned with a Type I or Type II error. Why?

Question 4

2 points

You are a sales agronomist with a grower who counts on you to bring them the latest, greatest opportunities to enhance yield. Their FOMO is pronounced. You are running a t-test on side-by-side data from a distributor, for a product which would cost \$3 per acre but has the potential to increase corn yield up to 4 bushels. Given the lower product cost and potentially high return on investment, as well as your concerns your customer would be upset at missing an opportunity to boost yield, are you more concerned about a Type I or Type II error in your analysis. Why?

Question 5

2 points

You are testing a new in-furrow fertilizer. It will cost more to produce, hence you will need to charge a greater market price. Thus, you will only release it if it produces significantly greater yield than your existing in-furrow product. You cannot afford to release a new product that does not work. Would you run a one-tailed or two-tailed test on your data?

Question 6

2 points

What would your null and alternative hypotheses be for the test in Question 5?

Question 7

3 points

Read in the dataset “data/wheat.csv”. The data set contains a column called “nitro” contains values “L” or “H”, meaning lower and higher nitrogen fertilization levels.

Conduct a one-tailed *t*-test where the high nitrogen (nitro=“H”) treatment yields **greater** than the low nitrogen treatment.

```
wheat <- read.csv('./data/wheat.csv')
t.test(yield ~ nitro,
       wheat,
       paired = TRUE,
       alternative = 'greater')
```

```
##
## Paired t-test
##
## data:  yield by nitro
## t = 6.0965, df = 83, p-value = 1.639e-08
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
##  39.68174      Inf
## sample estimates:
## mean difference
##      54.57143
```